White Paper

Advanced High Availability (AHA) Architecture

Delivering performance, scalability, and availability with the ServiceNow® AI Platform

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servicenow.

Introduction

This document provides an overview of the ServiceNow Advanced High Availability (AHA) architecture—a key element in delivering an enterprise-grade cloud service.

Organizations rely on access to IT, business data, and services for their continued operation and success. The ServiceNow multi-instance architecture meets and exceeds stringent requirements surrounding data sovereignty, availability, and performance.

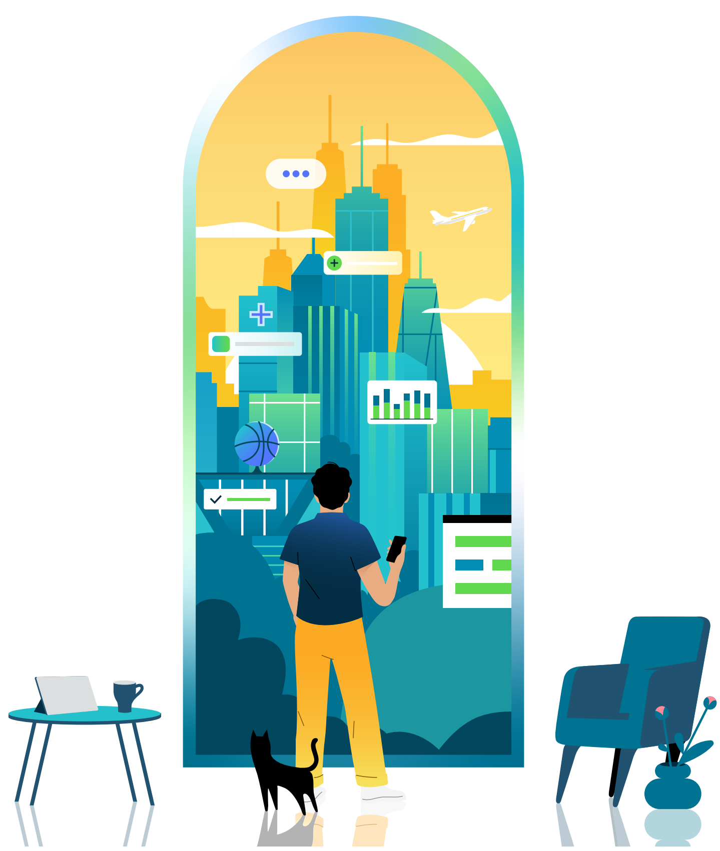
ServiceNow security

This document is designed to be read alongside Securing the ServiceNow AI Platform, which gives a holistic overview of the physical, administrative, and technical controls in place to secure the ServiceNow AI Platform, and how they combine to protect customer data.

Video: ServiceNow Availability and Resilience

This AHA white paper is available in other language versions: Spanish, French, German, Italian, Dutch, Brazilian Portuguese, Korean, and Japanese.

> *Note: All information in this white paper relates to the* **standard ServiceNow cloud**.
>
> *For information related to other globally located ServiceNow in-country cloud offerings and how these offerings may differ, please contact your ServiceNow account representative.*



White Paper

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Advanced High Availability (AHA) overview

Note: All information in this white paper is related to the **standard ServiceNow cloud**.

The ServiceNow Advanced High Availability (AHA) architecture is engineered to eliminate single points of failure across the ServiceNow AI Platform infrastructure. Each instance is backed by a multi-site network configuration with redundant Internet connections from multiple providers and redundant power sources, ensuring continuous availability at the site level.

At the core of AHA is an active-active site pairing model where all customer production data is hosted simultaneously across two geographically paired sites, kept in sync through continuous database replication. Both sites are always live and sized to independently carry the full production load, so in the event of a failure, failover can occur with minimal user interruption.

Within each site pair, there is no concept of a fixed primary site for any customer instance. For example, a customer with two separate instances could have their primary instance running out of different sites simultaneously.

ServiceNow leverages AHA for customer production instances for the following purposes:

- Prior to executing maintenance, ServiceNow can proactively transfer operation of a customer instance from one site to the other. The maintenance can then proceed without impacting service availability.
- In the event of failure of one or more infrastructure components, service is restored by transferring the operation of the affected instance to the other site.

With this approach, the transfer between active and standby sites is regularly executed as a standard operating procedure. This ensures that when it is needed to address a failure, the transfer will be successful, minimizing service disruption.

Transfer and failover

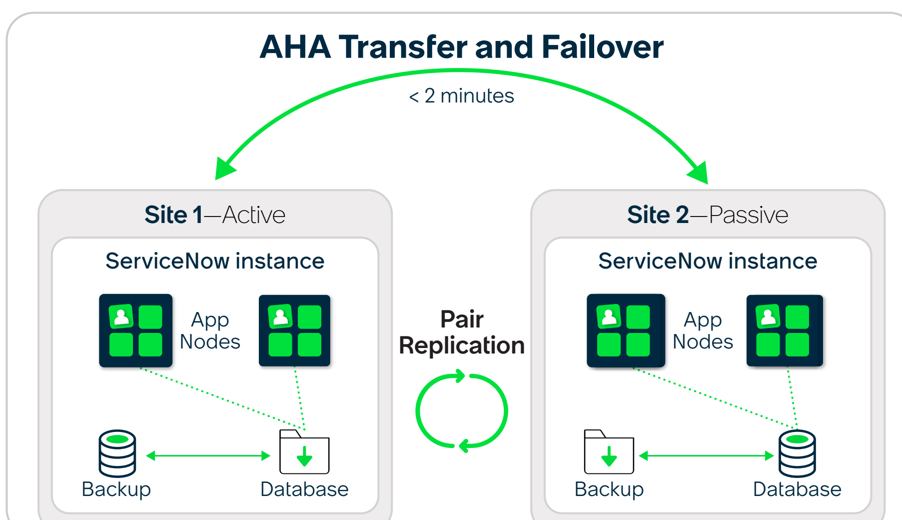
ServiceNow has two distinct processes related to ensuring instance availability: transfer and failover.

Transfer

A transfer of an instance is a planned event, usually performed for maintenance purposes. These outages occur within the contracted availability service level agreement that ServiceNow commits to with each customer.

Failover

A failover of an instance is an unplanned event usually performed when availability for one or more customer instances cannot be maintained. This could be caused by a local component failure, or an event such as a major environmental incident or resource outage.



In the case of a local component failure, a failover to a system within the same site will be attempted first. If a site-wide outage is identified, all current active production instances in the impacted site facility will be failed over to the alternate site in the pair. In this circumstance, a recovery time objective (RTO) of two hours, and a recovery point objective (RPO) of one hour is targeted.

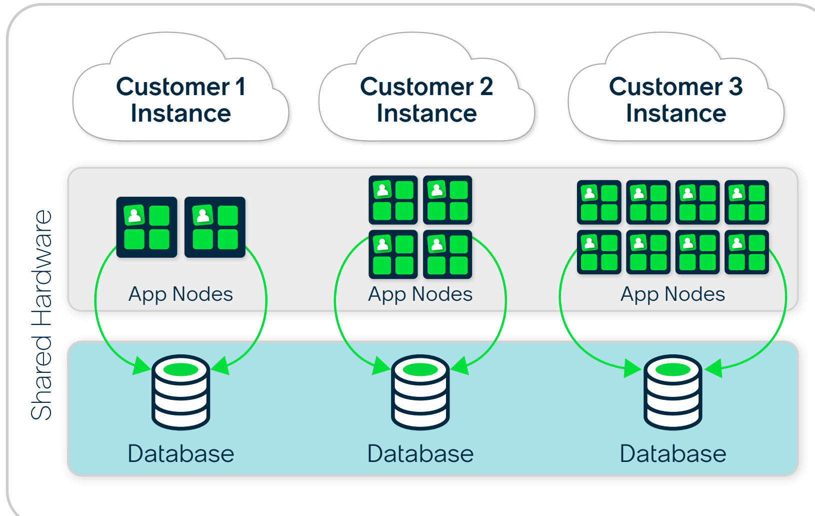
Automation technology built on the ServiceNow AI Platform is used to transfer or failover instances, when necessary. The mechanism for both processes is very similar. The current standby database is designated active, and vice versa. To complete the process, DNS mappings and instance database configurations are updated accordingly. Redundant DNS providers and DNSSEC (Domain Name System Security Extensions) are employed to provide robust, resilient name resolution services.

Multi-instance architecture

Instances of the ServiceNow AI Platform are deployed on an advanced, multi-instance architecture that provides separate application nodes and database processes for each customer.

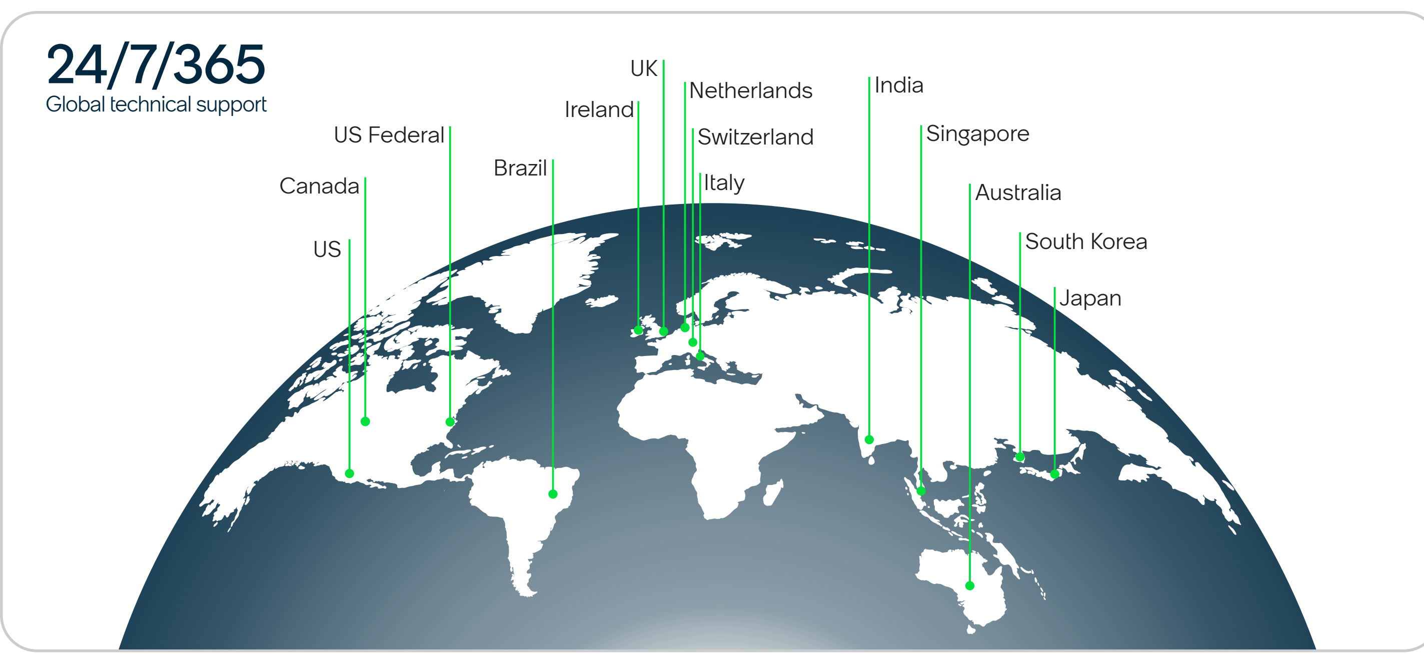
This cloud infrastructure ensures that one customer's data is processed separately from others and that any maintenance carried out on one customer's instance does not impact any other customer.

Because each ServiceNow AI Platform instance runs its own application logic and database processes, customers do not have to be on the same version or upgraded at the same time as other customers' instances. Therefore, customers can upgrade their instances on a schedule that best meets their needs and compliance requirements. No downtime is necessary for upgrades.



Global site pairs

Sites where the ServiceNow AI Platform is hosted are arranged in highly available site pairs, spanning five continents: Asia, Australia, Europe, North America, and South America.



ServiceNow partners with leading global infrastructure providers to support its operations. ServiceNow employs physical and logical controls where appropriate to restrict logical access to all ServiceNow infrastructure from providers.

Physical access to site locations is restricted to ServiceNow personnel who hold direct responsibilities or play an active role in maintaining physical infrastructure.

ServiceNow also offers site pairs exclusively for qualified US Federal and Swiss banking customers. Meeting regulatory and sovereignty obligations is a significant factor when selecting sites within specific geographic boundaries.

Performance and scalability

The ServiceNow cloud can scale to meet the needs of the largest global enterprises, with tens of thousands of customer instances operating in globally distributed sites. All instances are deployed per customer, allowing the multi-instance cloud to scale horizontally to meet each customer's performance needs.

Customer instances perform an aggregate of tens of billions of full-page transactions every month, and customers using the ServiceNow Configuration Management Database (CMDB) as their single system of record may manage tens of millions of Configuration Items (CIs).

Critical resources

ServiceNow is responsible for managing the environment, the supporting infrastructure, and vendor relationships. The ServiceNow Site Reliability Engineering (SRE) center employs a follow-the-sun model that provides continual operational monitoring and support of the ServiceNow environment and infrastructure. ServiceNow rotates operations and technical support daily in North America, the UK, India, Australia, the Netherlands, Japan, and Ireland.

Critical system resources, including DNS, email, ServiceNow cloud operation systems, and customer service systems, use high-availability configurations in at least two sites. These resources are independent of the internal ServiceNow corporate IT infrastructure.

ServiceNow uses AHA for development systems, including managing source code control and the software build process, which are also hosted at the production sites. The ServiceNow AHA architecture ensures the highest continuity for ServiceNow developers, enabling them to develop and support the application without requiring physical access to ServiceNow offices.

Data backup and recovery

The ServiceNow AHA architecture is the primary means to restore service in the case of a disruption that could impact availability.

Full backups are performed direct to disk every seven days, with differential backups performed every 24 hours. These backups are retained in accordance with ServiceNow standard operating procedures.

Backups are stored in the same sites where the data resides, with production instances backed up in both sites in the pair. Sub production instances (commonly used for testing and development purposes) are backed up only in their primary site, since they are not AHA-capable.

The ServiceNow backup architecture is not designed to provide archival records. However, customers may retain data within their instances for as long as they need during the subscription term, in accordance with their policy or regulatory requirements. Additionally, there are capabilities available within the ServiceNow AI Platform to allow customers to manage their unique instance logs and regularly export data to external systems, as required.

In specific scenarios, using more traditional data backup and recovery mechanisms may be appropriate. For example, if a customer deletes some data inadvertently or if a customer's data integration or automation gets misconfigured or malfunctions, this could render data unusable or inaccessible. In these scenarios, the high availability architecture would not be applicable, and restoring from backup is the only option for recovery.

Critical system resources, including DNS, email, ServiceNow cloud operation systems, and customer service systems, use high-availability configurations in at least two sites.

Conclusion

ServiceNow is committed to providing cloud services that are always highly available, with built-in redundancy across all network and server infrastructure.

All customer instances are individually provisioned on advanced multi-instance architecture, which ensures that one customer's data is processed separately from others and that any maintenance carried out on one customer's instance does not impact any other customer.

ServiceNow sites are arranged in highly-available pairs providing near-instant transfer from one site to another. This ensures Advanced High Availability for all customer production instances.

Upgrades can be performed at a schedule determined by each individual customer, with no downtime required. Full backups are performed every seven days, with differential backups taken every 24 hours.

Finally, the ServiceNow cloud can easily be scaled horizontally to meet the needs of even the largest global enterprises.

For further details on how ServiceNow delivers secure, scalable, and compliant cloud services, visit the ServiceNow Trust site.

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Resources

Videos: ServiceNow Security

- Securing the ServiceNow AI Platform white paper
- ServiceNow Security Best Practices Guide
- Cloud Security Customer Resources
- ServiceNow Product Documentation